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SEC-A

1. $\hookrightarrow$ Data and a pointer to the next node.
2. $\hookrightarrow$ NULL.
3. $\hookrightarrow$ self-referential structure.
4. $\hookrightarrow$ head $\rightarrow$ NULL
5. $\hookrightarrow$ newnode $\rightarrow$ Next $\rightarrow$ head; head = new node.
6. $\hookrightarrow$ temp = head; head = head $\rightarrow$ next; (free temp);
7. $\hookrightarrow$ traversing from tail to head.
8. $\hookrightarrow$ n.
9. $\hookrightarrow$ copying the data of $P \rightarrow$ next into P , then deleting $P \rightarrow$ next.
10. $\hookrightarrow$ while ($P \neq$ NULL)
11. $\hookrightarrow$ $P \rightarrow$ next $\rightarrow$ NULL
12. $\hookrightarrow$ two pointers advancing at different rates.
13. $\hookrightarrow$ Three.
14. $\hookrightarrow$ Exactly one nodes.
15. $\hookrightarrow$ $P = (\text{struct node}^*) \text{malloc}(\text{sizeof}(\text{struct node}));$
16. $\hookrightarrow$ Avoid sp. case handling for insertion and deletion at the head.
17. $\hookrightarrow$ The first node.
18. $\hookrightarrow$ It does not support random (index) access.

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19. (b) Loss of access to all nodes causing a memory leak.
20. (b) A slow pointer and a fast pointer.

21. (c) The last node.

22. (b) Nodes may be scattered anywhere in memory.

~~23. (a) (b) (c) (d)~~



Struct node {
int data;

Struct node *head;

};

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24. Output:

$$10 + 20 = 30$$

$$10 + 20 = 30$$

$$20 + 30 = 50$$

$$20 + 30 = 50$$

$$40 +$$

$$30 + 40 = 70$$

25. Error: if (*head == NULL) {

*head = t;

return;

}

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26. Struct node *reverse (Struct node *head) {

Struct node *prev = NULL;

Struct node *curr = head;

Struct node *next;

while (curr != NULL) {

next = curr -> next;

curr -> next = prev;

prev = curr;

curr = next;

return prev;

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28.

i) ~~7~~ 7 → 5 → 8 → 12 → 3 → NULL.

ii) 5 → 8 → 3 → NULL.

iii) 5 → 8 → ~~3~~ 12 → 3 → 9 → NULL.